Republic of the Philippines
Region V (Bicol)
BICOL UNIVERSITY - POLANGUI
Polangui, Albay

Fundamental of Electronics Circuits

LABORATORY 1 REPORT

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INTRODUCTION

The experiment investigated Ohm's Law and Kirchhoff's Law through real-world applications to understand its effect on actual resistors. The primary goal of the report is to compare the calculated resistances to the measured resistances of the resistors, demonstrating the accuracy of the calculations relative to the real values. This was essential for understanding the behavior of real-world resistors, which is relevant to the students' Bachelor of Science in Computer Engineering program. By conducting this experiment, the hypothesis that the real values would fall within the calculated range of possible values was tested. The methodology involved assembling circuits containing resistors connected in series, parallel, and series-parallel configurations. Depending on the parameter to be determined, either the voltage or the resistance of each resistor was measured and calculated. The color coding of each resistor was used for the calculated values, while a multimeter was employed for the measured values. Through this process, the students not only gained an understanding of how Ohm's Law applies to real-world resistors but also learned how to use a multimeter and determine the value of a resistor by interpreting the colored bands present.

LABORATORY #1 : RESISTANCE

I. OBJECTIVES

1. Learn to identify and interpret resistor color codes.
2. Accurately measure the resistance of a resistor using an ohmmeter.
3. Build series, parallel, and series-parallel circuits on a breadboard.
4. Calculate and measure the total resistance in different circuit configurations.

II. MATERIALS

MATERIAL	QUANTITY
Breadboard	1
Jumper Wires	2
Digital Multimeter	1
1k Ohms Resistor	1
330 Ohms Resistor	1
100 Ohms Resistor	1
47 Ohms Resistor	1

III. PROCEDURE

Using the breadboard, resistors, and jumper wires, 3 circuits were constructed. The first circuit was a series circuit, the second was parallel, and the third was series-parallel. Before constructing the circuits, each resistor was labeled as R_1, R_2, R_3, and R_4 then their resistance values were identified using their coloured bands. After constructing each circuit, the values of each resistor was measured using the multimeter. It is worth noting that the resistances, both measured and calculated, were recorded immediately after the values were identified. After the resistance of each resistor was identified, the total resistance of each circuit was calculated and measured. All gathered data was tabulated to present the data.

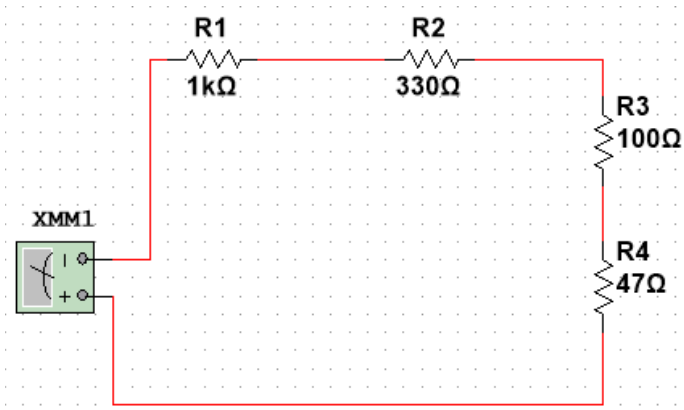

Figure 1. Series Circuit Diagram

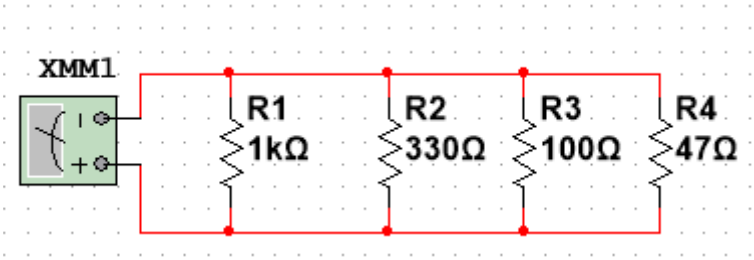

Figure 2. Parallel Circuit Diagram

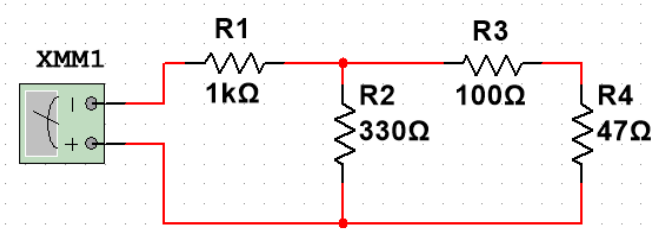

Figure 3. Series-Parallel Circuit Diagram

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IV. DATA

Resistor	Colours					Value	Range of possible values	Measured resistance
	1st band	2nd band	3rd band	4th band	5th band			
R_1 = 1kΩ	Brown	Black	Black	Brown	Brown	1000Ω ± 1%	Min: 990Ω Max:1010Ω	982Ω
R_2 = 330Ω	Orange	Orange	Black	Black	Brown	330Ω ± 1%	Min: 326Ω Max:333Ω	326Ω
R_3 = 100Ω	Brown	Black	Black	Black	Brown	100Ω ± 1%	Min: 99Ω Max:101Ω	99.6Ω
R_4 = 47Ω	Yellow	Purple	Black	Gold	Brown	47Ω ± 1%	Min: 46.53Ω Max:47.47Ω	46.5Ω

COMPUTED VS MEASURED DATA		
SERIES	COMPUTED TOTAL RESISTANCE :	1454.1 Ω
	MEASURED TOTAL RESISTANCE :	1456 Ω
PARALLEL	COMPUTED TOTAL RESISTANCE :	27.992 Ω
	MEASURED TOTAL RESISTANCE :	28.9 Ω
SERIES-PARALLEL	COMPUTED TOTAL RESISTANCE :	1082.887 Ω
	MEASURED TOTAL RESISTANCE :	1083 Ω

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LABORATORY #2 : VOLTAGE

I. OBJECTIVES

1. Measure the individual voltage drops of each resistor using voltmeter
2. Calculate the voltage drops using the identified Resistance values from each resistor and the known voltage input
3. Validate the measured voltages to the computed voltage drops

II. MATERIALS

MATERIAL	QUANTITY
Breadboard	1
Jumper Wires	2
Digital Multimeter	1
1k Ohms Resistor	1
330 Ohms Resistor	1
100 Ohms Resistor	1
47 Ohms Resistor	1

III. PROCEDURE

Continuing from Circuit 3, a power source was connected to bridge the open line. With this, the voltage drop can now be measured and calculated. To measure the voltage drop across each resistor, the multimeter was simply set to DC Voltage and the leads were attached across each component. After that, the values shown in the multimeter were recorded. As for the calculated voltages, Kirchhoff's Law was used in order to get the Voltages of each resistor. The working equations are as follow:

$$V_t = R_1 I_1 + R_2(I_1 - I_2)$$

$$R_2(I_1 - I_2) = R_3 I_2 + R_4 I_2$$

$$I_2 = \frac{R_1 I_1 + R_2 I_1 - V_t}{R_2}$$

$$I_1 = \frac{R_3 I_2 + R_4 I_2 + R_2 I_2}{R_2}$$

$$V_1 = R_1 I_1$$

$$V_2 = R_2(I_1 - I_2)$$

$$V_3 = R_3 I_2$$

$$V_4 = R_4 I_2$$

With all of these, the measured and calculated values can be tabulated and then compared.

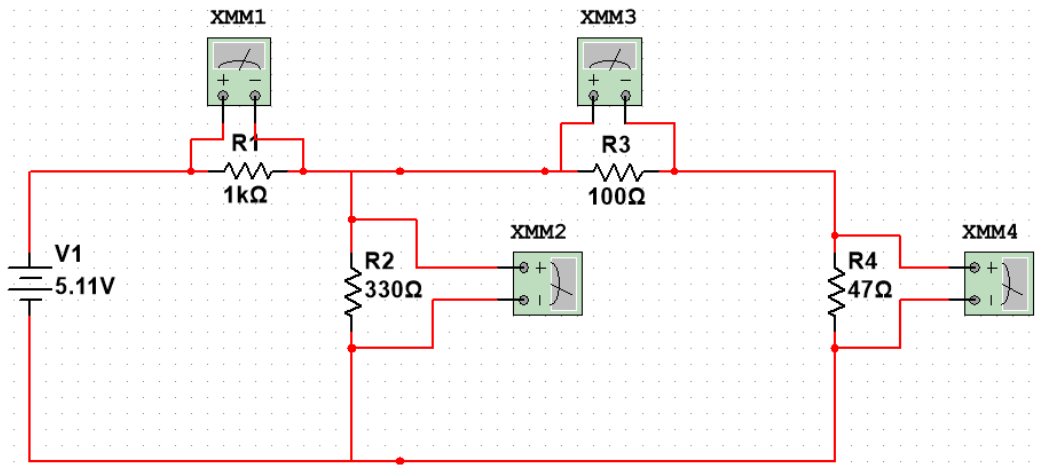

Figure 4. Powered Series-Parallel Circuit

IV. DATA

Resistor	Measured Voltage	Computed Voltage
V_1	4.63 V	4.635 V
V_2	0.47 V	0.476 V
V_3	0.31 V	0.325 V
V_4	0.14 V	0.151 V
V_T	5.11 V	5.111 V

CONCLUSION

The experiment has shown that the mathematics in Ohm's Law and Kirchhoff's Law works in the real world. The measured Resistance and Voltage in the circuits were almost identical to the calculated. This is evident in the Voltage Experiment. However, On the Resistance Experiments, most of the measured values did not fall within the range of the calculated values. This is due to other factors affecting resistances such as temperature and resistance of the jumper wires and breadboard that were not accounted for in the current laboratory work. If an additional 5% is to be considered, then the calculated values do predict the measured values. Overall, these measured values do not stray too far from the range and can support the claim of the students. With this, the alternative hypothesis is accepted while the null is rejected.

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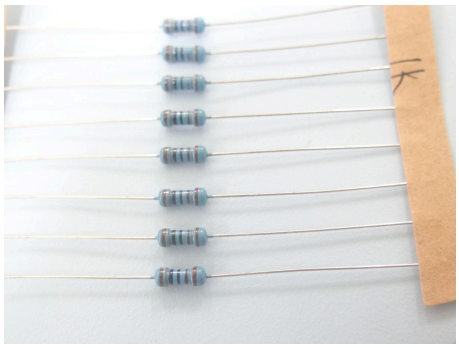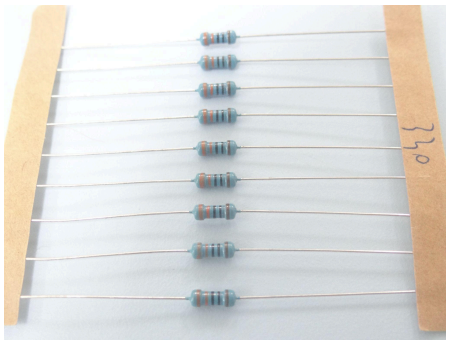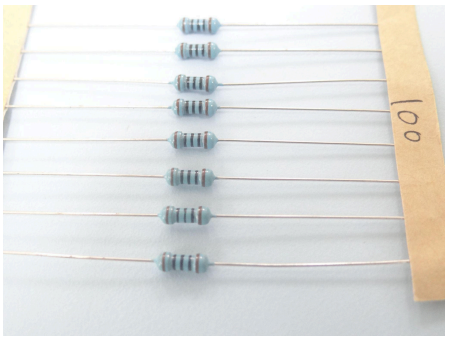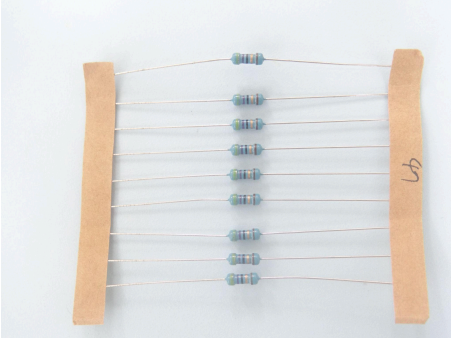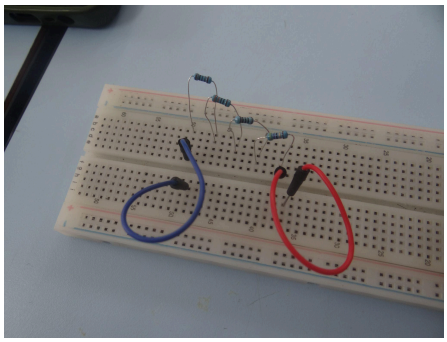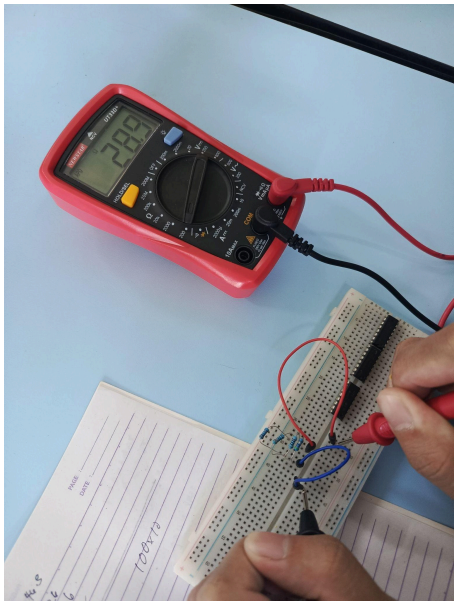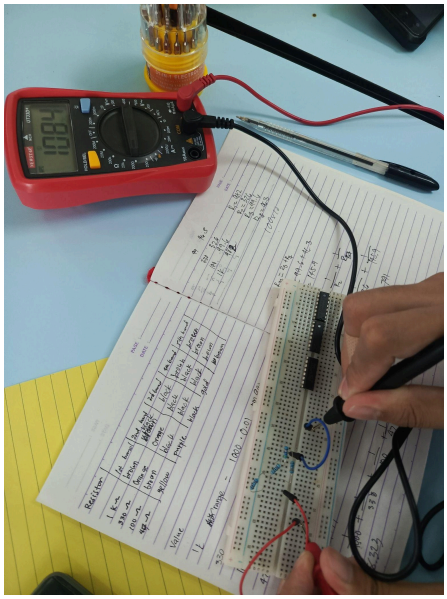
DOCUMENTATION